ADAPTIVE DIGITAL SAT PRACTICE TEST

SAT Practice Test 2 - Standardized Edition

Section 1: Reading and Writing

Module 1 (27 questions) | Module 2 Easier (27) | Module 2 Harder (27)

Section 2: Math

Module 1 (22 questions) | Module 2 Easier (22) | Module 2 Harder (22)

Total: 147 questions

Original test content retained. Module order and presentation standardized to the supplied reference structure.

Section 1: Reading and Writing

Module 1 (27 Questions) - All Students

Time: 32 minutes

Question 1

Skill: Words in Context | Difficulty: Easy

The Voyager space probes, launched in 1977, carry golden records containing sounds and images from Earth. Although originally designed for a five-year mission to study Jupiter and Saturn, the probes have far exceeded expectations. Scientist Suzanne Dodd, who manages the Voyager Interstellar Mission, continues to analyze transmissions from the probes, confirming that Voyager's contribution to our understanding of deep space is _______.

Which choice completes the text with the most logical and precise word or phrase?

A) debatable

B) unprecedented

C) ongoing

D) peripheral

Question 2

Skill: Words in Context | Difficulty: Easy

The following text is from Khaled Hosseini's 2003 novel *The Kite Runner*. The narrator describes learning to play chess as a child.

For the tournament, I was supposed to study a book of openings my father had bought me. It was a thick, serious volume with diagrams and annotations. I was meant to memorize the Sicilian Defense and the Queen's Gambit. But I skipped ahead, glancing at the diagrams without understanding them, and then improvising, moving pieces wherever they seemed right. I never truly grasped the strategies. I daydreamed about flying kites, about being somewhere above the rooftops.

Based on the text, when the narrator describes himself as "improvising," what does he most likely mean?

A) He was violating an expectation about how to prepare for the tournament.

B) He was gaining an unfair advantage over the other competitors.

C) He was deceiving his chess instructor.

D) He was lying to himself about his strategic ability.

Question 3

Skill: Words in Context | Difficulty: Easy

The Oxford English Dictionary's Reading Programme is a volunteer-driven initiative begun in the late nineteenth century to document the history of English words. Participants read published works and submit quotations illustrating word usage; these _______ from personal libraries and archives allow lexicographers to trace how terms such as "robot" (1920) first entered the language.

Which choice completes the text with the most logical and precise word or phrase?

A) negotiations

B) resolutions

C) contributions

D) justifications

Question 4

Skill: Words in Context | Difficulty: Easy

Modern concert halls are typically engineered with sophisticated acoustic panels and adjustable sound systems to optimize the audience's listening experience. Because concertgoers have become accustomed to polished acoustics, performances in raw, untreated spaces can startle listeners. But bare-walled venues were likely ________ audiences centuries ago: purpose-built concert halls with advanced acoustic design were not widespread until the 1800s.

Which choice completes the text with the most logical and precise word or phrase?

A) disliked by

B) confusing to

C) exciting to

D) expected by

Question 5

Skill: Main Purpose | Difficulty: Easy

Struggling with an overpopulation of deer? One surprising approach may be to introduce predators back into the ecosystem. Gray wolves and mountain lions are two native predators whose reintroduction into certain U.S. regions has helped control deer herds. These apex predators are sometimes more effective at managing populations than are traditional methods such as regulated hunting.

Which choice best states the main purpose of the text?

A) To describe how dangerous some predators are

B) To discuss the best way to manage wildlife

C) To suggest an unexpected solution to a problem

D) To identify a new ecological trend

Question 6

Skill: Overall Structure | Difficulty: Medium

Beloved, the 1987 novel by Ohio-born author Toni Morrison, is characteristic of African American literary fiction written during the 1970s and 1980s. During that era, Morrison and her contemporaries crafted haunting, psychologically complex narratives rooted in the historical trauma of slavery. Recently, however, younger Black writers have gravitated toward commercially popular genres known for being fast-paced and inventive, such as science fiction and Afrofuturism. Nigerian American author Nnedi Okofor is a prominent voice in this movement. Her 2010 novel *Who Fears Death* is a work of speculative fiction.

Which choice best describes the overall structure of the text?

A) It discusses fiction by earlier African American writers, then describes how fiction by more recent African American writers differs from the earlier fiction.

B) It praises one African American fiction writer, then criticizes a different African American fiction writer.

C) It provides an overview of African American fiction, then compares it to fiction by non-Black writers.

D) It describes one African American author's early works, then describes her more recent works.

Question 7

Skill: Function (Underlined Portion) | Difficulty: Medium

In 2016, marine biologist Danielle Dixson and colleagues published a study concluding that rising ocean temperatures significantly alter the migratory behavior of *Amphiprion ocellaris*, a species of clownfish. However, Dixson's study relied on a sample of only 12 fish from a single reef system. In a 2021 meta-analysis examining multiple researchers' conclusions about temperature effects on reef fish behavior, Frederick Bayer and colleagues cautioned that depending on such limited data sets can heighten the risk of skewed results. Such analysis, in turn, can lead to overstated conclusions.

Which choice best describes the function of the underlined sentence in the text?

A) It presents a criticism of the methodology reported by Dixson and colleagues in their 2016 study.

B) It lists several traits of *Amphiprion ocellaris* discovered by Dixson and colleagues while conducting their 2016 study.

C) It emphasizes a detail about where Dixson and colleagues conducted their 2016 study.

D) It states the conclusion reached by Dixson and colleagues in their 2016 study.

Question 8

Skill: Inference / Complete the Text | Difficulty: Medium

The atmosphere of Venus makes direct exploration of the planet exceptionally difficult, as the dense carbon dioxide clouds and surface temperatures exceeding 450 degrees Celsius are known to corrode sensors and disable electronic components on landers and other exploration equipment. Using analogs, which are materials engineered to replicate the atmospheric and surface conditions of other planets, scientists are able to study characteristics of Venus's environment. Analogs like the Venus Chamber Simulant (which was developed using sulfuric acid vapor and extreme heat in laboratory settings) help researchers evaluate how well their instruments will perform under Venusian conditions.

Based on the text, what is one reason why analogs are valuable for scientists?

A) Analogs allow scientists to test the ability of research equipment to withstand some of the conditions it will encounter during a mission.

B) Scientists use analogs to track how the chemical properties of planetary atmospheres have changed over time.

C) Analogs can be combined with materials from Earth to explore how research equipment will handle extreme environments on Earth.

D) Scientists use analogs to compare the physical properties of Venus's surface to those of Earth's surface.

Question 9

Skill: Evidence - Data / Graph / Table | Difficulty: Medium

GRAPH TO INSERT

[Table: Millions of Metric Tons of Lithium Extracted in 2000 and 2022. Australia: 2000 = 5.20, 2022 = 61.00. Chile: 2000 = 5.50, 2022 = 39.00. China: 2000 = 2.80, 2022 = 19.00. Argentina: 2000 = 1.20, 2022 = 6.20]

While doing research for a paper about battery minerals, a student finds information about lithium extraction in different countries in 2000 and 2022. The student notes that Australia produced 5.20 million metric tons of lithium in 2000 and _______.

A) 39.00 million metric tons of lithium in 2022.

B) 6.20 million metric tons of lithium in 2022.

C) 19.00 million metric tons of lithium in 2022.

D) 61.00 million metric tons of lithium in 2022.

Question 10

Skill: Function (Underlined Portion) | Difficulty: Medium

For decades, scholars attributed the earliest known use of the zero as a placeholder to Indian mathematician Brahmagupta in 628 CE. However, historian Agathe Keller has argued that Cambodian stone inscriptions from the Khmer Empire, predating Brahmagupta by roughly a century, contain a clear numeric symbol functioning as zero. <u>Keller asserts that in his role as an epigrapher, researcher Coedes identified a dot used as a placeholder in a Khmer inscription dated to 683 CE, and these records contain the earliest datable example of the zero symbol.</u>

Which finding, if true, would most directly support the underlined claim?

A) Several dots appear in both Brahmagupta's texts and Khmer inscriptions, but the dot was commonly used in manuscripts in the 600s and 700s as a means to separate words from numbers.

B) In a Khmer stone inscription from approximately 683 CE, a dot appears between two numerals in a context that clearly indicates a positional value of nothing.

C) In a table in Brahmagupta's astronomical text from the 620s, a small circle appears between two numbers in a context that suggests the value is not a whole number.

D) As recorded in Keller's research, some of Coedes's readings of Khmer inscriptions are inaccurate due to the use of whole numbers rather than placeholders.

Question 11

Skill: Evidence - Data / Graph / Table | Difficulty: Medium

GRAPH TO INSERT

[Table: Average Monthly Sick Days per Employee at Two Points After Wellness Programs Began. YGA (Yoga and Meditation): 6 Months = 3.1, 18 Months = 1.8. NTR (Nutrition Seminars): 6 Months = 3.3, 18 Months = 4.9]

Dr. Rachel Simmons et al. hypothesized that introducing wellness initiatives into offices would reduce absenteeism (employees missing work) and boost morale. Simmons's team enrolled groups of corporate workers in two programs: one that provided yoga and meditation sessions (YGA) and one that offered nutrition seminars (NTR). They then measured average monthly sick days at 6 months and 18 months after the programs started. They concluded that yoga-based wellness programs were more effective at reducing absenteeism than nutrition seminars, though this result took time to become apparent.

Which choice best describes data from the table that most effectively strengthens Simmons and colleagues' conclusion?

A) Absenteeism was largely due to low morale for the NTR group at 18 months, while it was due to illness for the YGA group at 18 months.

B) Sick days for the YGA group barely changed between 6 and 18 months, while sick days for the NTR group significantly increased during the same period.

C) Sick days were consistently higher for the NTR group over the entire measurement period, though the gap between the two narrowed over time.

D) Sick days were fairly similar for the YGA and NTR groups at 6 months, but at 18 months they had significantly increased for the NTR group and significantly decreased for the YGA group.

Question 12

Skill: Evidence - Data / Graph / Table | Difficulty: Medium

GRAPH TO INSERT

[Graph: Bar chart titled "Taste Rating: Labeled vs. Unlabeled" showing average taste ratings (1=lowest, 8=highest) for six foods: "Kale Chips," "Seaweed Snack," "Cricket Flour Bar," "Mealworm Pasta," "Spirulina Smoothie," and "Jackfruit Taco." Each food has two bars: light gray for "unlabeled" and dark gray for "labeled." For all six foods, the labeled bar is taller than the unlabeled bar. The gap is largest for "Cricket Flour Bar" and "Mealworm Pasta" (roughly 3-point difference) and smallest for "Kale Chips" and "Jackfruit Taco" (roughly 0.5-point difference).]

Researchers investigated how knowing what an unfamiliar food contains affects people's taste ratings. Participants rated foods once with ingredient labels visible and once without labels. For every food tested, participants who saw labels gave higher taste ratings than those who did not. But the magnitude of this labeling effect varied, as is best illustrated by the taste ratings for _______.

A) "Kale Chips" and "Jackfruit Taco."

B) "Seaweed Snack" and "Mealworm Pasta."

C) "Cricket Flour Bar" and "Mealworm Pasta."

D) "Spirulina Smoothie" and "Jackfruit Taco."

Question 13

Skill: Inference / Complete the Text | Difficulty: Hard

Cats are overwhelmingly right-paw dominant (73 to 80%) and can thus be said to exhibit strong population-level right-pawedness (PLRP). Among studies of non-domestic felines, Maria K. Llorente's 2011 study of captive lions purported to show PLRP, while Craig Packer's 1986 study of wild lions did not. Overall, the studies claiming PLRP in non-domestic felines find much lower incidences of right-pawedness than domestic cats exhibit, but it's worth noting that studies of captive felines tend to show significantly greater incidences of right-pawedness than studies of wild felines do, therefore raising the possibility that _______.

Which choice most logically completes the text?

A) the number of individuals in the study of wild lions is insufficient to preclude the claim that non-domestic felines exhibit PLRP.

B) the number of individuals in the study of captive lions is insufficient for a robust claim regarding evidence of PLRP.

C) the greater exposure to humans among the captive lions induced them to acquire more right-pawed behaviors.

D) the apparent discrepancy between the studies' results may be partly attributable to the 1986 study using different criteria when assessing paw dominance than were used in the 2011 study.

Question 14

Skill: Inference / Complete the Text | Difficulty: Hard

The red-crowned crane and the white-naped crane are large migratory birds that winter in wetlands such as the Poyang Lake region in China. Keisuke Ueda and colleagues wanted to understand how these birds select overwintering areas. They examined habitat characteristics including water depth and the duration of seasonal flooding. They found that although only red-crowned cranes prefer areas with shallow water during winter, both red-crowned cranes and white-naped cranes prefer areas that experience seasonal flooding for more than 90 days per year. The researchers therefore concluded that neither species is very drawn to areas where _______.

Which choice most logically completes the text?

A) there is seasonal flooding for far fewer than 90 days per year.

B) species with overwintering seasons longer than 90 days are likely to be present.

C) there are any features that attract the other species.

D) there is relatively shallow water during the overwintering season.

Question 15

Skill: Inference / Complete the Text | Difficulty: Hard

It is challenging for publishers and printers to produce very long books that balance the demands of readability, durability, and affordability. Among some exceptionally long novels published in English, Leo Tolstoy's *War and Peace* (translated from Russian) was published in a 1,225-page edition and runs to an estimated 580,000 words, while Donna Tartt's *The Goldfinch* (written originally in English) was published in a 771-page edition and runs to an estimated 297,000 words. This indicates that _______.

Which choice most logically completes the text?

A) the edition of *War and Peace* has more words per page than the edition of *The Goldfinch*.

B) translating a text from Russian to English is likely to increase the word count.

C) page count is likely to be a more reliable indicator of how long it takes to read a book than word count.

D) the edition of *War and Peace* is printed on thinner paper than the edition of *The Goldfinch*.

Question 16

Skill: Grammar / Conventions | Difficulty: Medium

Girl with a Pearl Earring is one of the most recognized works by the Dutch Golden Age painter Johannes Vermeer. _______ around 1665, the painting can now be found at the Mauritshuis museum in The Hague.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) Will complete

B) Completed

C) Completes

D) Was completed

Question 17

Skill: Grammar / Conventions | Difficulty: Medium

Engineered titanium alloy implants (TAIs) can be used to repair fractured bones and reinforce weakened joints; this capability, well suited for numerous orthopedic procedures, _______ from the exceptional strength-to-weight ratio of these alloys.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) stemming

B) having stemmed

C) stems

D) to stem

Question 18

Skill: Grammar / Conventions | Difficulty: Medium

The Sino-Tibetan language family comprises more than 400 _______ Mandarin and Cantonese, which are spoken by over 1 billion and 85 million speakers, respectively—and accounts for a significant portion of the world's linguistic diversity, making it of great interest to researchers like George van Driem.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) languages
B) languages—
C) languages:
D) languages,

Question 19

Skill: Grammar / Conventions | Difficulty: Medium

For the past 50 million years, the continents have been drifting toward their current positions at a relatively steady pace. This has not always been the _______ throughout geologic history, the continents have collided and separated multiple times through a process called the supercontinent cycle.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) case; though,
B) case, though,
C) case, though;
D) case, though

Question 20

Skill: Grammar / Conventions | Difficulty: Hard

Japanese-born visual artist Yayoi Kusama transforms gallery spaces into immersive, otherworldly environments. In her installations, mirrored walls create infinite reflections of glowing orbs, polka-dotted sculptures fill entire _______ suspended nets of light shift and shimmer overhead. This "infinity room" concept, Kusama says, "is about erasing the boundaries of the self."

Which choice completes the text so that it conforms to the conventions of Standard English?

A) rooms,
B) rooms—
C) rooms;
D) rooms:

Question 21

Skill: Transitions | Difficulty: Medium

Eighteenth-century Rococo architects championed ornamentation in their designs. Granted, the lavish gilding and curved moldings of the Amalienburg pavilion, a Rococo hunting lodge designed by François Cuilliés, couldn't exactly pass for a natural setting. _______ one sees organic influences in Cuilliés's preference for flowing curves over rigid geometry and his use of leaf and shell motifs.

Which choice completes the text with the most logical transition?

A) Furthermore,
B) Still,

C) Similarly,

D) In other words,

Question 22

Skill: Transitions | Difficulty: Medium

In Japan, citizens must be at least 18 years old to vote in national elections. ________ citizens in Scotland can vote in local elections at the age of 16.

Which choice completes the text with the most logical transition?

A) Firstly,

B) For example,

C) By comparison,

D) Therefore,

Question 23

Skill: Transitions | Difficulty: Medium

Generally, compact vehicles are more fuel-efficient than larger ones. For example, the aerodynamic profile of the Toyota Prius helps it achieve exceptional mileage. ________ a full-size pickup truck encounters greater air resistance, making it less fuel-efficient.

Which choice completes the text with the most logical transition?

A) Specifically,

B) On the other hand,

C) As a result,

D) In conclusion,

Question 24

Skill: Transitions | Difficulty: Hard

As an abolitionist, Boston minister William Ellery Channing opposed slavery and advocated for human dignity and moral reform; ________ sermons he delivered at Federal Street Church in the 1830s forcefully challenged the arguments of pro-slavery advocates who defended the institution on economic grounds.

Which choice completes the text with the most logical transition?

A) in other words,

B) by comparison,

C) fittingly,

D) nevertheless,

Question 25

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes:

- Tulum is a municipality in the state of Quintana Roo, Mexico.
- Municipalities are governmental regions responsible for providing many public services to their residents.
- One service they provide is waste management.

• Tulum covers an area of roughly 2,090 km².
• Quintana Roo is divided into 11 municipalities.

The student wants to emphasize the size of Tulum. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) The municipality of Tulum in Quintana Roo, Mexico, covers an area of roughly 2,090 km².

B) Tulum, a governmental region in the state of Quintana Roo, Mexico, provides many public services to its residents.

C) Tulum is one of 11 governmental regions, known as municipalities, across Quintana Roo.

D) Providing waste management is just one example of the public services that municipalities provide.

Question 26

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes:
• Canada has designated more than 80 areas as National Marine Conservation Areas (NMCAs).
• Some NMCAs were established specifically to protect endangered marine species.
• The Gwaii Haanas NMCA is a 3,400 km² area in British Columbia.
• It was established to protect endangered Pacific glass sponge reefs.
• The Fathom Five NMCA is a 112 km² area in Ontario.
• It was established to protect endangered freshwater shipwreck ecosystems.

The student wants to emphasize a similarity between the two NMCAs. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Both the Gwaii Haanas NMCA and the Fathom Five NMCA were established to protect endangered species or ecosystems.

B) Some NMCAs, such as Ontario's Fathom Five, were established specifically to protect endangered ecosystems.

C) While the Gwaii Haanas NMCA extends across 3,400 km², the Fathom Five NMCA encompasses only 112 km².

D) Canada has designated more than 80 areas as NMCAs, including the Gwaii Haanas NMCA in British Columbia.

Question 27

Skill: Rhetorical Synthesis | Difficulty: Hard

While researching a topic, a student has taken the following notes:
• Takashi Murakami (pseudonym Mr. DOB) is a Tokyo-based artist who specializes in the form of art known as Superflat.
• Blurring the boundary between commercial and fine art is a central aspect of Superflat.
• Playful mixing of traditional Japanese art with pop culture imagery is a central aspect of Superflat.
• Murakami's *Tan Tan Bo Puking* playfully mixes elements of classical Rinpa painting, manga, and corporate branding.
• Murakami: "My art is a journey through the border between high and low, a place where tradition and pop culture collide and transform each other."

The student wants to provide a specific example of playful mixing in Superflat art. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Elements of classical Rinpa painting, manga, and corporate branding are playfully mixed in the Superflat piece *Tan Tan Bo Puking* by artist Takashi Murakami.

B) The playful mixing of traditional and pop elements in Takashi Murakami's art evokes "a place where tradition and pop culture collide."

C) *Tan Tan Bo Puking* by artist Takashi Murakami is an example of Superflat, a form in which playful mixing is a central aspect.

D) Superflat art, such as Takashi Murakami's *Tan Tan Bo Puking*, blurs boundaries between commercial and fine art and playfully mixes disparate elements.

Section 1: Reading and Writing

Module 2 - Easier Path (27 Questions)

Time: 32 minutes

Question 1

Skill: Words in Context | Difficulty: Easy

Like other barrier islands, the Outer Banks of North Carolina is _______: it is a constantly shifting chain of narrow sandy islands that change in width and elevation as hurricanes and ocean currents redistribute sediment along the Atlantic coastline.

Which choice completes the text with the most logical and precise word or phrase?

A) unrivaled
B) dynamic
C) sustainable
D) immutable

Question 2

Skill: Words in Context | Difficulty: Easy

Noticing the similarity between Frida Kahlo's *Self-Portrait with Thorn Necklace* and her husband Diego Rivera's mural style, art historian Teresa del Conde argued that the former reflects Rivera's influence: that by incorporating bold colors and symbolic imagery reminiscent of Rivera's murals, Kahlo _______ a shared artistic vision. Indeed, by digitally analyzing both works, del Conde _______ the two pieces' stylistic parallels.

Which choice completes the text with the most logical and precise word or phrase?

A) corroborated
B) amalgamated
C) promulgated
D) consolidated

Question 3

Skill: Words in Context | Difficulty: Medium

While contemporary architects may often draw inspiration from past masters like Frank Lloyd Wright, the breadth of their influences is wide. The designs of Zaha Hadid, for instance, can be said to _______ elements of both classical mathematical geometry and the organic flowing forms of the natural landscape.

Which choice completes the text with the most logical and precise word or phrase?

A) defy
B) neglect
C) foretell
D) exhibit

Question 4

Skill: Words in Context | Difficulty: Hard

During the 2015 to 2016 refugee crisis, Germany accepted over one million asylum seekers from conflict zones. Political scientist Hanne Fjelde found that regional governments' responses to immigration seem to have been _______ local integration policies: the likelihood that municipalities would establish refugee support programs was 0.45 in regions with proactive outreach and 0.08 in regions with restrictive frameworks.

Which choice completes the text with the most logical and precise word or phrase?

A) predicated on

B) mediated by

C) decoupled from

D) material to

Question 5

Skill: Main Purpose | Difficulty: Medium

Though Ursula K. Le Guin, author of *The Left Hand of Darkness*, is perhaps not as widely known as the most commercially successful science fiction writers of the past half century, influential figures have championed her work, including the novelist Margaret Atwood and the literary critic Harold Bloom. In his introduction to Le Guin's novel *The Dispossessed*, Bloom praises the book's masterful synthesis of what philosopher Herbert Marcuse called the utopian and critical modes: while Marcuse believed works could be classified as either imagining ideal societies (utopian) or exposing existing flaws (critical), *The Dispossessed* demonstrates that a work can be both.

Which choice best states the main purpose of the text?

A) To present a reason why a literary critic is impressed by a certain novel

B) To explain what inspired an author to write a particular work

C) To compare the work of a writer with the work of a philosopher who admired her

D) To argue that all writing must be classified as belonging to one of two categories

Question 6

Skill: Overall Structure | Difficulty: Hard

The following text is adapted from Charles Dickens's 1853 novel *Bleak House*. Mr. Turveydrop is a dancing master in the city of London.

By the merchants of the neighborhood and its vicinity [Mr. Turveydrop] was regarded as a man of quite remarkable poise and elegance, who must present a magnificent figure in the best ballrooms of the metropolis, and by his fellow dancing masters he was esteemed as a courteous and accomplished gentleman. Mr. Turveydrop never entered into an argument with passion; he implied what others might consider, but seldom declared what he believed himself; he never permitted anyone to detect that he was performing a role, and he never gave any person cause to question his sincerity.

Which choice best describes the overall structure of the text?

A) It implies that Mr. Turveydrop's neighbors are more naive in their estimation of him than people in London are and then explains why his neighbors have been so easily impressed.

B) It presents the favorable opinion of Mr. Turveydrop that other people hold and then describes the behaviors of Mr. Turveydrop that enable him to maintain that favorable opinion.

C) It shows that Mr. Turveydrop had originally been held in high regard by his colleagues and then details the events that caused their regard for him to subside.

D) It highlights the disparity between Mr. Turveydrop's public and private behavior and then conveys why he labors to obscure his true self from other people.

Question 7

Skill: Cross-Text | Difficulty: Hard

Text 1 is adapted from Virginia Woolf's 1925 novel *Mrs Dalloway*. Text 2 discusses *Mrs Dalloway*. The Strand and Bond Street are adjacent shopping streets in London through which the protagonist walks.

Text 1

To Clarissa the pavement of Bond Street had always suggested permanence. Its very stones, withdrawn a little from the roaring Strand, implied a serenity beyond the grasp of commerce. Those dignified shop fronts, unchanging, indifferent, shouldering between them a quiet clock, were fit monuments to some enduring quality, whose expression would certainly not be found in the language of shopkeeping.

Text 2

The tension between opposing states of mind in *Mrs Dalloway* is articulated through the novel's geography of London streets. On one hand, the busy Strand represents modern life's noise, speed, and commercial energy; on the other, Bond Street, accessible from the Strand, embodies an elegance that values beauty, permanence, and contemplation, or what the novel treats as "the enduring."

Based on the texts, the author of Text 2 would most likely agree with which statement about Bond Street, as it is depicted in Text 1?

A) Because it is situated adjacent to the busy Strand, Bond Street symbolizes the encroachment of commercial culture into spaces the novel associates with beauty and reflection.

B) The quietness conveyed by Bond Street's appearance mirrors Clarissa's dissatisfaction with the prospect of finding authentic meaning in a world that primarily values consumption.

C) As a point of contrast with the Strand, Bond Street suggests to Clarissa the possibility of experiencing the lasting beauty promised by the contemplative spaces the novel seems to favor.

D) Bond Street has a relatively modest appearance whose sharp contrast with the more energetic atmosphere of the Strand suggests to Clarissa the dominance of commercial culture among Londoners.

Question 8

Skill: Inference / Complete the Text | Difficulty: Hard

Brazilian architect Oscar Niemeyer's prolific career, which spanned the 1930s to the 2000s, evolved through distinct phases. After traveling to Europe in the early 1930s and immersing himself in Le Corbusier's modernist philosophy, Niemeyer began incorporating sweeping curves and open spaces derived from European modernism in his work, as seen in the Pampulha complex in Belo Horizonte, whose fluid organic forms contrast with his earlier projects in Rio de Janeiro, such as the Santos Dumont Airport, which reflect the rectilinear aesthetics of traditional Brazilian Neoclassicism.

Information in the text best supports which statement about the design of the Santos Dumont Airport?

A) It represents a transitional moment between the early and late phases of Niemeyer's development.

B) It displays the effects of Niemeyer's exposure to international architectural trends in the 1930s.

C) It is characteristic of the Rio de Janeiro architecture that influenced Niemeyer throughout his career.

D) It reflects an approach to form and shape that Niemeyer later moved away from.

Question 9

Skill: Inference / Complete the Text | Difficulty: Hard

The following text is from Hilary Mantel's 2009 novel *Wolf Hall*. The narrator is describing a member of the king's court.

At last, and providing a good deal of what one might call ballast, came Sir Thomas Boleyn, whose expression seemed designed to ornament a monument, or perhaps a ship's prow, and whose brow was so conspicuously furrowed with calculation that it would have been discourteous to examine too carefully the actual fruits his three decades of diplomatic service had yielded. His recently issued memoirs might have illuminated this puzzle, but since their substance did not emerge until well past the opening chapters, they might just as well have been sealed correspondence.

Based on the text, which choice best describes Sir Thomas Boleyn?

A) He has the appearance of a distinguished figure, but it is uncertain whether he has accomplished anything to earn distinction.

B) He has maintained a modest profile even though he has served the crown capably for many years.

C) He looks like a person worthy of respect, but his memoirs reveal that some of his actions were dishonorable.

D) He would be a celebrated public figure if his achievements did not have to be kept secret.

Question 10

Skill: Evidence - Support / Weaken | Difficulty: Hard

Glasgow, Scotland, has installed green infrastructure along 72% of its riverbanks to reduce flooding and improve water quality, a practice known as riverbank naturalization. To evaluate the responses of urban bird populations to two types of naturalization (native wildflower meadows and constructed wetlands), Dr. Fiona Campbell et al. surveyed songbird communities consisting of robins, blackbirds, and 45 other species at different sites along the River Clyde. Using the Index of Songbird Community Health (ISCH), on which a high score corresponds to high community health, the researchers found that wildflower meadows are more strongly positively correlated with songbird community health than are constructed wetlands.

Which finding, if true, would most directly illustrate the researchers' finding?

A) Songbird communities at Riverside, a site with a high percentage of bank covered by meadows and wetlands, had lower average ISCH scores than did songbird communities at Parkhead, a site with a low percentage of bank covered by meadows and wetlands.

B) Songbird communities at Dalmarnock, a site with a relatively high percentage of bank consisting of meadows, had lower average ISCH scores than did songbird communities at Bridgeton, a site with a relatively high percentage of bank consisting of wetlands.

C) Songbird communities at Tollcross, a site with equal percentages of bank consisting of meadows and wetlands, had higher average ISCH scores than did songbird communities at Shettleston, a site with different percentages of meadows and wetlands.

D) The difference in average ISCH scores for songbird communities at Govan and Partick, two sites with a higher percentage of bank consisting of meadows than of wetlands, was statistically insignificant.

Question 11

Skill: Quotation Illustrates Claim | Difficulty: Hard

For its 2023 exhibition *Behind the Lens*, the National Museum of Photography invited the security guards typically responsible for protecting the museum's collections to serve as curators, whose individual selections from the museum's archives culminated in a show <u>amplifying a distinct perspective not typically given a formal platform</u>.

Which quote from one of the personnel who curated the exhibition would best support the underlined claim?

A) "The most enjoyable part of curating this show has been seeing the outcome of our efforts: a unique show that spans eras, techniques, and subjects. We hope visitors will sense our affection for the pieces we are entrusted with guarding."

B) "My selections for the show were motivated by an interest in bringing attention to lesser-known photographs in the museum's collection, many of which tend to be passed over or even dismissed as unremarkable by most visitors to the museum."

C) "The photograph I chose is accompanied by a statement of the personal interpretation I've developed over countless hours standing beside the work and observing visitors' reactions, an interpretation that I've previously shared only in small impromptu conversations with those visitors who pause to discuss the piece with me."

D) "Throughout the exhibition's development, my colleagues and I partnered closely with other museum staff whose areas of expertise lie in curation and archival preservation, and whose insights enriched our understandings of the pieces we selected for the exhibition."

Question 12

Skill: Rhetorical Synthesis | Difficulty: Hard

A student is writing an essay on the subject of vertical farming, which involves growing crops indoors in stacked layers and is intended to help reduce the environmental impact of traditional agriculture. The student wants to make the case that vertical farming may be useful in responding to an expected rise in global food demand.

Which quotation from a publication by a researcher would most effectively support the student's claim?

A) "Consumers tend to believe that reducing pesticide use in indoor farming would have a significant effect on food safety."

B) "The taste of hydroponically grown produce differs across crop varieties and is also influenced by lighting conditions and nutrient solutions."

C) "A growing population that is adding substantially more protein to its diet will contribute to an increasing demand for food production in the 21st century."

D) "Researchers who advocate for the expansion of vertical farming claim that it is better for the environment than conventional agriculture because it requires less arable land."

Question 13

Skill: Inference / Complete the Text | Difficulty: Hard

Born in Argentina in 1921, composer Astor Piazzolla was a pioneer of the nuevo tango (New Tango) movement that emerged in the 1950s and then spread throughout Latin America, Europe, and Japan as nuevo tango. Piazzolla traveled across Buenos Aires and beyond, studying classical composition techniques as well as traditional milonga rhythms, folk melodies, and other facets of Argentine musical heritage. These studies formed the foundation for the early movement's fusion of traditional Argentine tango forms with jazz harmonies and classical structures in new compositions that represented modern urban life and pushed artistic boundaries. <u>As the movement spread beyond Argentina, the range of</u>

musical traditions woven into its fabric also expanded.

Which detail about songs associated with nuevo tango, if true, would best illustrate the underlined claim?

A) Many feature rhythmic patterns addressing contemporary social issues that stemmed from shared experiences of political upheaval in South American countries.

B) Many demonstrate the stylistic influence of bossa nova, a genre of Brazilian music that had come to be characterized by lyrical melodies in the early 1960s.

C) Many were written with parts meant to be played on the bandoneon, a traditional instrument used across South American countries, including Argentina.

D) Many were produced by Japanese artists in the late 1970s, with others by artists in additional countries first emerging soon after.

Question 14

Skill: Inference / Complete the Text | Difficulty: Hard

The Clean Air Act requires US federal agencies to evaluate the air-quality effects of proposed industrial activities, such as building power plants. While many Clean Air Act assessments require public hearings, general conformity determinations (GCDs) allow streamlined reviews without hearings for activities that will have a minimal effect on air quality. In 2021 the rule governing GCDs was revised: before, GCDs could be issued for actions that "do not individually or cumulatively have a significant effect on air quality," but the revised rule allowed GCDs if actions "normally do not have a significant effect on air quality." Environmental groups found this revision to be potentially harmful because _______.

Which choice most logically completes the text?

A) the increasing need to build new industrial facilities in the US incentivizes agencies to issue GCDs after 2021 for reasons that would not have been considered valid prior to 2021.

B) the 2021 relaxation of the rule regarding GCDs would permit more streamlined reviews, resulting in more approvals by federal agencies and a paradoxically slower review process.

C) the rule that governed GCDs before 2021 allowed expedited reviews of actions that might have significant effects on air quality if those effects were believed to be rare, while the 2021 revision subjected such actions to slower reviews.

D) the 2021 revision of the rule governing GCDs would allow expedited reviews of actions that had a minimal air-quality effect when considered on their own but a significant effect when considered together.

Question 15

Skill: Inference / Complete the Text | Difficulty: Hard

Microbial electrolysis cells (MECs) harness the ability of certain bacteria to break down organic waste and produce hydrogen gas. The bacteria form colonies on the surface of an anode, but transferring the resulting electrons from the bacterial membrane to an external circuit requires the electrons to pass through a series of inefficient metabolic steps. Accordingly, MEC hydrogen output rarely exceeds a rate of 2.5 liters per day (L/d). In an experiment, researchers coated the anode with graphene nanoparticles. The resulting hydrogen output was 5.8 L/d. Since materials such as graphene exhibit exceptional electrical conductivity, the researchers hypothesized that _______.

Which choice most logically completes the text?

A) graphene nanoparticles may increase the metabolic processes of the bacteria, thereby increasing the number of electrons available for the circuit.

B) as the density of the bacterial colony increases, the metabolic steps may accelerate independent of the presence of the graphene nanoparticles.

C) electrons may be conducted directly to the circuit before the graphene nanoparticles catalyze the metabolic steps.

D) graphene nanoparticles may allow electrons to bypass the series of metabolic steps and transfer directly to the circuit.

Question 16

Skill: Grammar / Conventions | Difficulty: Medium

Featured in *Radical Fiber*, a 2019 group exhibition at the Textile Museum in Washington, D.C., was the work of artist Faith Ringgold, who is best known for her narrative quilts that juxtapose painted imagery with written stories. Her work challenges conventional notions of race, gender, history, and _______ she is credited with expanding the horizons of textile-based art.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) memory,

B) memory and

C) memory

D) memory, and

Question 17

Skill: Grammar / Conventions | Difficulty: Medium

Ratified in 2010, Kenya's constitution, according to legal _______ contains all of the eight constitutional features that promote governmental transparency. Explicit provisions for freedom of information, Odhiambo and Akech's research explains, are more likely to be found in constitutions enacted after 2000.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) researchers Odhiambo and Akech

B) researchers Odhiambo and Akech,

C) researchers, Odhiambo and Akech

D) researchers, Odhiambo and Akech,

Question 18

Skill: Grammar / Conventions | Difficulty: Medium

Although both 433 Eros and 25143 Itokawa are classified as near-Earth asteroids (rocky bodies that orbit close to our planet), they exhibit striking differences in _______ object 433 Eros is considered a monolithic asteroid with a solid, unfractured interior, while 25143 Itokawa, showing a loose collection of rubble held together by gravity, is considered a rubble-pile asteroid.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) behavior. The

B) behavior; while the

C) behavior, while the

D) behavior; the

Question 19

Skill: Grammar / Conventions | Difficulty: Medium

In a parliamentary election, one might expect major parties to focus on the most densely populated districts. However, if polls and historical voting patterns suggest that the outcome in a given district is already settled, a party may choose not to campaign there. Ultimately, a district's electoral history and polling data, not its population size, _______ its strategic importance to campaigns.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) has determined

B) determines

C) determining

D) determine

Question 20

Skill: Grammar / Conventions | Difficulty: Hard

That the repeated eruptions of Mount Etna, a stratovolcano on the island of Sicily, have occurred on timescales as brief as six months (shorter intervals between eruptions than have been observed in some other volcanoes of its class) _______ to the volcano's position above an exceptionally active subduction zone and rapid magma supply from its mantle source.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) is due

B) due

C) being due

D) having been due

Question 21

Skill: Grammar / Conventions | Difficulty: Hard

In her 2015 book, sociologist Arlie Russell Hochschild documents the political attitudes of working-class communities in rural Louisiana. Hochschild's ethnographic study _______ nearly 60 in-depth interviews conducted between 2011 and 2016 reveals not only persistent economic anxieties but also a deepening distrust of federal institutions, particularly those related to environmental regulation.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) had examined

B) examining

C) examines

D) examined

Question 22

Skill: Transitions | Difficulty: Medium

Scientists studying asteroid deflection have focused on secondary objects such as S/2020 (2013 PY6), a moonlet orbiting the near-Earth asteroid 2013 PY6. In 2022 NASA intentionally crashed a probe into just such an object, the moonlet Dimorphos. _______ Dimorphos's orbital period around the near-Earth asteroid Didymos was permanently altered.

Which choice completes the text with the most logical transition?

A) In comparison,

B) In addition,

C) Consequently,

D) Specifically,

Question 23

Skill: Transitions | Difficulty: Medium

According to the band theory of solids, metals such as gold and aluminum do not form bonds through directional sharing of electrons. _______ metals are held together by a sea of delocalized electrons that move freely, interacting with neither specific atoms nor each other. This framework for understanding the electronic structure within metals can be used to help explain thermal and electrical conductivity.

Which choice completes the text with the most logical transition?

A) For example,

B) Rather,

C) Accordingly,

D) Likewise,

Question 24

Skill: Transitions | Difficulty: Hard

In an ecphrastic poem, a poet explores a work of visual art, such as a drawing, sculpture, or painting. John Keats's 1819 poem "Ode on a Grecian Urn," for example, meditates on an ancient Greek vase; _______ Tracy K. Smith's 2011 poem "Hubble Photographs" takes as its subject a series of deep-space images captured by the Hubble Space Telescope.

Which choice completes the text with the most logical transition?

A) nevertheless,

B) conversely,

C) likewise,

D) hence,

Question 25

Skill: Transitions | Difficulty: Hard

Honeybee phospholipase (HbPLA) is a distinctly evolved enzyme that can cleave a substrate called phosphatidylcholine to release anti-inflammatory compounds and can also act as an antimicrobial peptide, a class of protein present in all insects. _______ HbPLA is a bifunctional enzyme whose presence indicates an insect capable of producing anti-inflammatory compounds; in contrast, the presence of antimicrobial peptides alone would be insufficient for determining this capability.

Which choice completes the text with the most logical transition?

A) Nevertheless,

B) In fact,

C) Moreover,

D) That is,

Question 26

Skill: Rhetorical Synthesis | Difficulty: Medium

A student has gathered the following information:

• The Bangladesh Garment Workers' Federation (BGWF) is a national labor union for the South Asian nation of Bangladesh.

• It advocates for improved working conditions and safety standards for garment workers.

• IndustriALL Global Union is an international federation that represents national labor unions across manufacturing sectors.

• It represents the BGWF.

Which choice most effectively uses information from the given sentences to explain the purpose of national union federations?

A) The BGWF is a national labor union in Bangladesh.

B) National labor unions, such as the BGWF, work to improve conditions for workers in their member nations.

C) The BGWF is one of the national labor unions represented by IndustriALL Global Union.

D) International federations like IndustriALL represent the national labor unions in a specific sector.

Question 27

Skill: Rhetorical Synthesis | Difficulty: Hard

While researching a topic, a student has taken the following notes:

• Generally, a material will expand when heated.

• The expansion of a material when heated is known as thermal expansion.

• A 2020 study led by Hiroshi Yamamoto and Kenji Takahashi tested the thermal expansion of various polymers.

• When a sample of polyethylene (PE) plastic was heated, its average length increased by 0.12 mm.

• When a sample of polytetrafluoroethylene (PTFE) was heated, its average length increased by 0.98 mm.

The student wants to emphasize a similarity between PE and PTFE. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) In 2020, two research teams observed the effects of thermal expansion on various polymers, including PE and PTFE.

B) Both PE and PTFE expand when heated, according to a 2020 study.

C) Researchers determined that when heated, the average length of PTFE increased more than that of PE.

D) Heating a material will generally cause it to expand, a process known as thermal expansion.

Section 1: Reading and Writing

Module 2 - Harder Path (27 Questions)

Time: 32 minutes

Question 1

Skill: Words in Context | Difficulty: Hard

In her 2018 study of Renaissance-era painters, art historian Giovanna Ferretti argues that the attribution of anonymous works to canonical masters is often _______: while connoisseurs may confidently assign a painting to Raphael or Titian on the basis of stylistic analysis, subsequent chemical dating of pigments has frequently contradicted these assessments, exposing a persistent gap between subjective expertise and empirical verification.

Which choice completes the text with the most logical and precise word or phrase?

A) inconsequential
B) tenuous
C) indisputable
D) meticulous

Question 2

Skill: Words in Context | Difficulty: Hard

Certain deep-sea organisms, such as the giant tube worm *Riftia pachyptila*, have no digestive tract and instead rely on chemosynthetic bacteria housed in a specialized organ called the trophosome. This symbiotic arrangement is so tightly integrated that the worm's metabolism is entirely _______ the bacteria's ability to oxidize hydrogen sulfide, rendering the organism incapable of independent nutrient acquisition.

Which choice completes the text with the most logical and precise word or phrase?

A) contingent on
B) insulated from
C) tangential to
D) conducive to

Question 3

Skill: Words in Context | Difficulty: Hard

Economist Daron Acemoglu contends that institutions, not geography or culture, are the primary determinants of national prosperity. His thesis is that inclusive institutions (those that distribute political power broadly and provide economic incentives for innovation) _______ sustained growth, whereas extractive institutions (those that concentrate power and divert resources to elites) perpetuate poverty even in resource-rich nations.

Which choice completes the text with the most logical and precise word or phrase?

A) engender
B) preclude

C) circumvent

D) diminish

Question 4

Skill: Words in Context | Difficulty: Hard

In quantum mechanics, the principle of superposition holds that a particle can exist in multiple states simultaneously until it is measured, at which point the wave function collapses into a single outcome. This behavior, however counterintuitive, has been ________ by decades of experimental results, leaving little room for alternative interpretations that deny the phenomenon's reality.

Which choice completes the text with the most logical and precise word or phrase?

A) undermined

B) overstated

C) anticipated

D) substantiated

Question 5

Skill: Main Purpose | Difficulty: Hard

Literary scholar Saidiya Hartman's 2019 work *Wayward Lives, Beautiful Experiments* challenges conventional historical methodology by reconstructing the inner lives of young Black women in early twentieth-century Philadelphia and New York. Drawing on arrest records, case files, and institutional reports, Hartman does not merely extract facts from these documents; she uses them as springboards for speculative narrative, imagining what these women may have felt and desired. In doing so, she deliberately blurs the line between historical analysis and creative imagination, arguing that this hybrid approach is necessary because the traditional archive was designed to record the perspectives of those in power, not the perspectives of those who were surveilled, disciplined, and confined.

Which choice best states the main purpose of the text?

A) To explain why a scholar's unconventional methodology is suited to her subject matter

B) To critique the limitations of historical documents from early twentieth-century cities

C) To compare the effectiveness of speculative narrative with that of traditional historical analysis

D) To provide an overview of the lives of young Black women in Philadelphia and New York

Question 6

Skill: Overall Structure | Difficulty: Hard

The Cambrian Explosion, which began roughly 541 million years ago, is often cited as the moment when complex animal life first appeared on Earth. Fossils from the Burgess Shale formation in British Columbia reveal an extraordinary variety of body plans, including creatures with five eyes and armored appendages that have no modern analogues. Some paleontologists have attributed this sudden diversification to rising oxygen levels, which would have supported the metabolic demands of larger, more complex organisms. Others, however, point to ecological factors, arguing that once the first predators evolved, an evolutionary "arms race" drove rapid morphological innovation as prey species developed increasingly elaborate defenses. A third hypothesis suggests that changes in developmental genes, particularly the Hox gene family, made it possible for organisms to develop novel body structures more easily.

Which choice best describes the overall structure of the text?

A) It introduces a research question and then describes the methods scientists used to investigate it.

B) It describes a scientific finding and then provides evidence that discredits earlier explanations of it.

C) It summarizes a debate and then argues that one hypothesis is more plausible than the others.

D) It identifies an event and then presents three different hypotheses that attempt to explain it.

Question 7

Skill: Cross-Text | Difficulty: Hard

Text 1

In a 2020 paper, developmental psychologist Lisa Chen argued that bilingual children's code-switching (alternating between languages within a single conversation) reflects sophisticated cognitive processing rather than linguistic confusion. Chen demonstrated that children as young as three strategically switch languages depending on their listener's language proficiency, suggesting that code-switching requires advanced awareness of social context.

Text 2

Sociolinguist Ravi Desai, in a 2022 study of bilingual communities in Gujarat, India, observed that code-switching among adults follows predictable patterns governed by the topic of conversation. Speakers tend to switch to English for discussions of technology and global commerce but revert to Gujarati for matters of family and tradition, indicating that code-switching is shaped by the cultural associations each language carries rather than by the social characteristics of the listener alone.

Based on the texts, how would Desai (Text 2) most likely respond to Chen's argument in Text 1?

A) By noting that Chen's emphasis on listener awareness captures only one of several factors that influence code-switching decisions

B) By arguing that Chen's findings apply exclusively to children and therefore cannot be generalized to adult bilingual speakers

C) By agreeing with Chen's claim and providing additional evidence from adult speakers that supports the same conclusion

D) By questioning whether the children in Chen's study were truly bilingual or merely had passive familiarity with a second language

Question 8

Skill: Inference / Complete the Text | Difficulty: Hard

The following text is from Fyodor Dostoevsky's 1866 novel *Crime and Punishment* (translated by Constance Garnett). Raskolnikov has just committed a crime and returned to his apartment.

He sat down on the sofa, feeling that he could not stand for another instant. A fit of shivering came over him. Mechanically he drew towards him an old student's cloak, which was lying on a chair beside him, and covered himself with it. He was not asleep and not awake. Then a thought flashed through his mind, and he started: he remembered that he had left the door unlocked. He would have leaped up, but his limbs refused to obey him. He felt the cloak around him and tried to think clearly, but nothing came. He simply lay there, trembling.

Based on the text, which choice best describes Raskolnikov's psychological state?

A) He is exhausted from physical labor but otherwise calm and composed.

B) He is deliberately pretending to be incapacitated so that he will not be suspected of wrongdoing.

C) He is experiencing a conflict between the urgent need to act and his body's inability to comply.

D) He is gradually regaining control over his thoughts after a period of confusion.

Question 9

Skill: Evidence - Data / Graph / Table | Difficulty: Hard

GRAPH TO INSERT

[Table: Average Sleep Quality Scores (10-Point Scale) Before and After 8-Week Light Exposure Study. Natural Light Group: Before = 5.1, After = 7.5 (Change = +2.4). Artificial Light Group: Before = 5.3, After = 5.6 (Change = +0.3). Control Group (No Intervention): Before = 5.2, After = 5.1 (Change = –0.1).]

Which choice most effectively uses data from the table to support the researchers' conclusion that exposure to natural light has a greater effect on sleep quality than exposure to artificial light?

A) Participants in the natural light group reported an average improvement of 2.4 points on a 10-point sleep quality scale, while participants in the artificial light group reported an average improvement of 0.3 points.

B) Participants in the natural light group slept an average of 7.8 hours per night, which is within the recommended range for adults, while participants in the artificial light group slept an average of 7.2 hours per night.

C) The natural light group had a dropout rate of 5%, compared to a dropout rate of 12% in the artificial light group.

D) Both groups reported higher sleep quality scores at the end of the study than at the beginning, with the natural light group's scores increasing by 2.4 points and the artificial light group's scores increasing by 1.9 points.

Question 10

Skill: Evidence - Data / Graph / Table | Difficulty: Hard

GRAPH TO INSERT

[Graph: A scatterplot showing GDP per capita (x-axis, in thousands of dollars) versus average life expectancy (y-axis, in years) for 120 countries. A logarithmic trend line is overlaid. Most data points cluster near the trend line, but several countries with GDP per capita between $10,000 and $40,000 show a steady, linear increase rather than the expected plateau.]

A researcher claims that the relationship between a country's GDP per capita and its average life expectancy follows a logarithmic pattern: life expectancy increases sharply at lower GDP levels but plateaus at higher GDP levels. Which observation from the graph would most directly weaken this claim?

A) Several countries with GDP per capita above $50,000 have life expectancies that are lower than those of countries with GDP per capita between $20,000 and $30,000.

B) The data points for countries with GDP per capita below $5,000 show a wide range of life expectancies, from 45 to 70 years.

C) Countries with GDP per capita between $10,000 and $40,000 show a steady, linear increase in life expectancy with no sign of leveling off.

D) Two outlier countries with very high GDP per capita have the highest life expectancies in the data set.

Question 11

Skill: Evidence - Support / Weaken | Difficulty: Hard

Archaeologist Elena Vasquez has proposed that the decline of the ancient Harappan civilization around 1900 BCE was primarily caused by a shift in the monsoon pattern that reduced rainfall in the Indus Valley, leading to agricultural collapse. Vasquez points to pollen records from lake sediments that show a transition from crop-associated species to drought-resistant grasses around that period.

Which finding, if true, would most directly weaken Vasquez's hypothesis?

A) Archaeological excavations at Mohenjo-daro reveal that many Harappan cities were abandoned gradually over a period of roughly 200 years, rather than suddenly.

B) Isotopic analysis of Harappan-era cattle bones indicates that the animals' diets shifted from grain-based feed to wild grasses, consistent with declining agricultural productivity.

C) Hydrological modeling shows that the Ghaggar-Hakra river system, which supplied water to many Harappan settlements, changed course around 2000 BCE due to tectonic activity rather than reduced rainfall.

D) Trade records from Mesopotamian sites indicate a decline in imports from the Indus Valley beginning around 1900 BCE.

Question 12

Skill: Quotation Illustrates Claim | Difficulty: Hard

Political theorist Hannah Arendt, in her 1963 work *Eichmann in Jerusalem*, introduced the concept of "the banality of evil" to describe how ordinary individuals can participate in atrocities without recognizing the moral gravity of their actions. <u>Arendt argued that Eichmann's evil was not the product of fanaticism or sociopathy but of thoughtlessness, an inability or refusal to consider the consequences of his bureaucratic decisions from the perspective of those who suffered them.</u>

Which quotation from a scholar would best support the underlined claim?

A) "Arendt's analysis reveals that Eichmann was a zealous ideologue whose deep commitment to Nazi ideology motivated his participation in the Holocaust."

B) "Arendt's portrait of Eichmann suggests that moral catastrophe can arise not from extraordinary malice but from the ordinary human tendency to follow orders without reflection."

C) "Arendt's account emphasizes that Eichmann was fully aware of the suffering he caused but chose to suppress his empathy in service of his career ambitions."

D) "Arendt's concept of the banality of evil has been widely criticized for understating the role of antisemitic ideology in motivating perpetrators of the Holocaust."

Question 13

Skill: Inference / Complete the Text | Difficulty: Hard

Ecologist Suzanne Simard's research on forest ecosystems has revealed that trees in a forest do not merely compete for resources; they also share nutrients through underground fungal networks, sometimes called "wood wide web" connections. <u>Simard found that "mother trees" (large, older trees) preferentially send carbon and nutrients to their own seedlings, even when unrelated seedlings are also connected to the network.</u>

Which finding, if true, would most directly illustrate the underlined claim?

A) A Douglas fir seedling genetically related to a nearby mother tree received 40% more carbon through the fungal network than an unrelated seedling of the same species growing at the same distance from the mother tree.

B) Forests with intact fungal networks have 25% higher overall biomass than forests where the networks have been disrupted by logging.

C) Mother trees connected to fungal networks had higher survival rates during drought conditions than isolated trees of the same age.

D) Seedlings of all species growing near mother trees showed improved growth rates compared to seedlings growing in clearings.

Question 14

Skill: Inference / Complete the Text | Difficulty: Hard

The concept of "attention residue," introduced by business professor Sophie Leroy, describes a phenomenon in which a person's cognitive performance on a new task is impaired because their attention is still partially engaged with a previous, incomplete task. Leroy demonstrated that subjects who were allowed to finish one task before beginning another performed significantly better on the second task than subjects who were interrupted midway through the first task. This finding has implications for workplace design, suggesting that policies promoting uninterrupted blocks of focused work may improve overall productivity because _______.

Which choice most logically completes the text?

A) workers who are interrupted frequently develop coping mechanisms that allow them to switch between tasks more efficiently over time.

B) employees are less likely to leave residual attention on a previous task if they have had the opportunity to bring it to completion before transitioning.

C) the cognitive demands of most workplace tasks are insufficient to fully engage a worker's attention, leaving excess capacity that can be allocated to secondary tasks.

D) workers who report feeling more productive in uninterrupted environments also tend to report higher job satisfaction, which independently improves performance.

Question 15

Skill: Inference / Complete the Text | Difficulty: Hard

The Mpemba effect, in which hot water can freeze faster than cold water under certain conditions, has puzzled physicists for decades. One proposed explanation involves convection currents: hot water generates stronger convection patterns that can distribute heat more efficiently to the container's surface area, accelerating heat loss. However, this explanation predicts that the effect should disappear when convection is suppressed, such as in containers with narrow openings. Experiments conducted by physicist James Brownridge, using sealed containers that minimized convection, still observed the Mpemba effect, suggesting that _______.

Which choice most logically completes the text?

A) convection currents may play a role in some instances of the Mpemba effect but are not the sole mechanism responsible for the phenomenon.

B) the Mpemba effect occurs more reliably in containers with wide openings than in containers with narrow openings.

C) hot water freezes faster than cold water primarily because of differences in dissolved gas content rather than differences in temperature.

D) the containers Brownridge used were not sufficiently narrow to suppress convection entirely, invalidating his results.

Question 16

Skill: Grammar / Conventions | Difficulty: Hard

The axolotl, a neotenic salamander native to the lake complex underlying Mexico City, retains larval features throughout its life, a trait that has made it invaluable to regenerative medicine researchers. Unlike most amphibians, which undergo metamorphosis and lose the ability to regenerate complex tissues, the axolotl _______ full limbs, portions of its spinal cord, and even parts of its brain and heart.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) to regrow
B) regrowing
C) can regrow
D) having regrown

Question 17

Skill: Grammar / Conventions | Difficulty: Hard

The James Webb Space Telescope, launched in December 2021, has produced images of galaxies that formed within 300 million years of the Big Bang. These early galaxies, far more luminous and massive than existing models _______, have prompted astrophysicists to reconsider fundamental assumptions about the rate at which structures formed in the early universe.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) have predicted
B) predicting
C) will predict
D) had predicted

Question 18

Skill: Grammar / Conventions | Difficulty: Hard

The Japanese concept of *wabi-sabi*, which finds beauty in imperfection and transience, has influenced not only traditional art forms such as tea ceremony and pottery _______ contemporary Western architects, who have begun to incorporate deliberately imperfect materials and asymmetrical designs into their buildings.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) but also the work of
B) and also, the work of
C) but, also the work of
D) but also, the work of

Question 19

Skill: Grammar / Conventions | Difficulty: Hard

In 1928, Scottish bacteriologist Alexander Fleming noticed that a mold colony on an uncovered Petri dish had created a bacteria-free zone around _______ an observation that led to the discovery of penicillin and launched the era of antibiotics.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) itself,
B) itself:
C) itself;
D) itself

Question 20

Skill: Grammar / Conventions | Difficulty: Hard

The collection of rare first-edition books that the university library acquired last year, along with the accompanying archive of the author's unpublished correspondence, _______ now housed in a climate-controlled vault accessible only to approved researchers.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) are

B) is

C) were

D) have been

Question 21

Skill: Transitions | Difficulty: Hard

Cognitive scientist George Lakoff has argued that abstract thinking is largely structured by metaphor: people understand complex concepts like time, morality, and causation by mapping them onto concrete physical experiences such as spatial movement and bodily sensation. _______ when English speakers say that a deadline is "approaching" or that the past is "behind" them, they are drawing on a spatial metaphor that conceptualizes time as physical motion along a path.

Which choice completes the text with the most logical transition?

A) Nevertheless,

B) Accordingly,

C) In contrast,

D) For instance,

Question 22

Skill: Transitions | Difficulty: Hard

The 2011 Fukushima Daiichi nuclear disaster prompted Germany to accelerate its phase-out of nuclear power, shutting down its remaining reactors by April 2023. _______ France, which derives approximately 70% of its electricity from nuclear energy, has committed to building additional reactors, viewing nuclear power as essential to meeting its carbon reduction targets.

Which choice completes the text with the most logical transition?

A) Similarly,

B) Consequently,

C) In contrast,

D) Specifically,

Question 23

Skill: Transitions | Difficulty: Hard

Researchers at the University of Cambridge found that people who reported reading literary fiction scored significantly higher on tests of empathy than those who reported reading primarily nonfiction or genre fiction. The researchers attributed this difference to the demands that literary fiction places on readers: interpreting ambiguous characters and unreliable narrators requires sustained perspective-taking, a skill that _______ transfers to real-world social interactions.

Which choice completes the text with the most logical transition?

A) despite this,

B) for example,

C) as such,

D) evidently,

Question 24

Skill: Transitions | Difficulty: Hard

Traditional economic models assume that individuals make rational decisions aimed at maximizing their own utility. Behavioral economist Daniel Kahneman, however, demonstrated that people systematically deviate from rationality in predictable ways. _______ his research on loss aversion showed that individuals feel the pain of losing $100 approximately twice as strongly as they feel the pleasure of gaining $100, a bias that leads to risk-averse behavior even when the expected value of a gamble is positive.

Which choice completes the text with the most logical transition?

A) In summary,

B) Regardless,

C) In particular,

D) By contrast,

Question 25

Skill: Rhetorical Synthesis | Difficulty: Hard

While researching a topic, a student has taken the following notes:

• The Great Pacific Garbage Patch (GPGP) is a massive accumulation of ocean plastic debris.
• A 2018 study estimated the GPGP covers approximately 1.6 million square kilometers.
• Approximately 94% of the estimated 1.8 trillion pieces of plastic in the GPGP are microplastics (smaller than 5 mm).
• Despite their small size, microplastics are difficult to remove because they are dispersed throughout the water column rather than floating on the surface.
• The Ocean Cleanup project has developed a system designed to capture larger pieces of floating debris but acknowledges that microplastic removal remains a significant challenge.

The student wants to explain why the GPGP is difficult to clean up despite technological efforts. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) The GPGP, estimated to cover 1.6 million square kilometers, contains approximately 1.8 trillion pieces of plastic, 94% of which are microplastics.

B) Although the Ocean Cleanup project has developed systems to capture floating debris from the GPGP, 94% of the patch consists of microplastics dispersed throughout the water column, making comprehensive removal extremely difficult.

C) The Ocean Cleanup project, which targets the GPGP, has developed technology to capture larger pieces of floating debris from the ocean's surface.

D) Microplastics, which are smaller than 5 mm, make up approximately 94% of the plastic in the GPGP, which covers roughly 1.6 million square kilometers.

Question 26

Skill: Rhetorical Synthesis | Difficulty: Hard

While researching a topic, a student has taken the following notes:

• CRISPR-Cas9 is a gene-editing technology that allows scientists to make precise modifications to DNA sequences.

• In 2020, Jennifer Doudna and Emmanuelle Charpentier received the Nobel Prize in Chemistry for developing CRISPR-Cas9.

• One application of CRISPR-Cas9 is the development of disease-resistant crop varieties, which could improve global food security.

• Another application is gene therapy for genetic disorders such as sickle cell disease, where early clinical trials have shown promising results.

• Ethical concerns have been raised about using CRISPR-Cas9 for germline editing, which would make heritable changes to the human genome.

The student wants to highlight a contrast between the potential benefits and the ethical concerns associated with CRISPR-Cas9. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) CRISPR-Cas9, developed by Nobel laureates Jennifer Doudna and Emmanuelle Charpentier, has applications in agriculture and medicine, including the development of disease-resistant crops and gene therapy for sickle cell disease.

B) While CRISPR-Cas9 holds promise for improving food security and treating genetic disorders like sickle cell disease, its potential use for germline editing has raised ethical concerns about heritable changes to the human genome.

C) Ethical concerns about CRISPR-Cas9 focus primarily on germline editing, a process that would introduce heritable changes to human DNA.

D) CRISPR-Cas9 has both agricultural and medical applications: it can be used to create disease-resistant crops and to treat genetic disorders such as sickle cell disease.

Question 27

Skill: Rhetorical Synthesis | Difficulty: Hard

While researching a topic, a student has taken the following notes:

• The Svalbard Global Seed Vault is located on a Norwegian Arctic island.

• It stores backup copies of seeds from gene banks around the world to protect against the loss of crop diversity.

• As of 2024, the vault holds over 1.3 million seed samples from nearly every country.

• In 2015, researchers withdrew seeds from the vault for the first time to replace collections destroyed during the Syrian civil war.

• The vault's location was chosen because the permafrost and thick rock surrounding it provide natural freezing conditions even if the facility's mechanical cooling systems fail.

The student wants to provide evidence that the vault has already served its intended purpose. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) The Svalbard Global Seed Vault, located on a Norwegian Arctic island, holds over 1.3 million seed samples from gene banks in nearly every country.

B) The vault's remote Arctic location was selected because the surrounding permafrost and rock provide natural freezing that would preserve the seeds even without mechanical cooling.

C) The vault was designed to store backup copies of seeds so that lost collections could be replaced, and in 2015, seeds were withdrawn for the first time to restore gene bank collections destroyed during the

Syrian civil war.

D) The Svalbard Global Seed Vault protects crop diversity by storing seeds from gene banks around the world in a naturally cold Arctic environment.

Section 2: Math

Module 1 (22 Questions) - All Students

Time: 35 minutes

Question 1

Skill: Linear Equations in One & Two Variables | Difficulty: Easy

The equation $d = 25 + 3t$ gives the distance d, in miles, of a cyclist from a starting point t hours after the cyclist began riding. What is the distance, in miles, of the cyclist from the starting point 5 hours after the cyclist began riding?

A) 25
B) 37
C) 40
D) 50

Question 2

Skill: Triangle Properties | Difficulty: Easy

[Figure: A right triangle with hypotenuse labeled 25, one leg labeled 7, and the other leg labeled a. The right angle is at the vertex between legs 7 and a. Note: Figure not drawn to scale.]

Which equation shows the relationship between the side lengths of the given triangle?

A) $7a = 25$
B) $7 + a = 25$
C) $7^2 + a^2 = 25^2$
D) $7^2 - a^2 = 25^2$

Question 3

Skill: Data Interpretation from Tables and Graphs | Difficulty: Easy

[Figure: A histogram showing the distribution of 25 weights, in kilograms. First bar (0 to less than 10 kg): frequency 3. Second bar (10 to less than 20 kg): frequency 5. Third bar (20 to less than 30 kg): frequency 4. Fourth bar (30 to less than 40 kg): frequency 6. Fifth bar (40 to less than 50 kg): frequency 7.]

The histogram shows the distribution of 25 weights, in kilograms, in a data set. The first bar represents the weights that are less than 10 kilograms, the second bar represents the weights that are at least 10 kilograms but less than 20 kilograms, and so on. Which of the following could be the maximum weight, in kilograms, in this data set?

A) 9
B) 29
C) 39
D) 47

Question 4

Skill: Linear Equations in One & Two Variables | Difficulty: Easy

If $3x + 12 = 27$, what is the value of $x + 4$?

A) 5
B) 9
C) 15
D) 19

Question 5

Skill: Systems of Equations - Number of Solutions | Difficulty: Easy

$9x + 35 = 9x + k$

In the given equation, k is a constant. The equation has infinitely many solutions. What is the value of k?

Enter your answer as a number.

Question 6

Skill: Linear Equations in One & Two Variables | Difficulty: Easy

A rope that has a length of 156 centimeters (cm) is cut into three pieces. One piece is 52 cm long. The other two pieces have lengths that are equal to each other. What is the length, in cm, of one of the other two pieces of equal length?

Enter your answer as a number.

Question 7

Skill: Linear Equations in One & Two Variables | Difficulty: Easy

A store sells notebooks for $4 each and pens for $2 each. If a customer buys a total of 15 items and spends $42, how many notebooks did the customer buy?

Enter your answer as a number.

Question 8

Skill: Coordinate Geometry | Difficulty: Medium

[Figure: A scatterplot showing the relationship between two variables, x and y. A line of best fit is also shown. The line passes through approximately (–12, 16) and (0, 4), with a negative slope. Points are scattered around the line.]

Which of the following equations best represents the line of best fit shown?

A) $y = x – 18.5$
B) $y = –x + 18.5$
C) $y = x – 4.0$
D) $y = –x + 4.0$

Question 9

Skill: Polynomial Operations | Difficulty: Medium

The expression $5x^4 + 11x^4 – 9x^4$ is equivalent to bx^4, where b is a constant. What is the value of b?

Enter your answer as a number.

Question 10

Skill: Systems of Linear Equations | Difficulty: Medium

At what point (x, y) do the graphs of the equations $y = 3x + 5$ and $y = 4x – 3$ intersect in the xy-plane?

A) (5, 3)
B) (3, 5)
C) (8, 29)
D) (29, 8)

Question 11

Skill: Quadratic Word Problems | Difficulty: Medium

A rectangle has a length that is 45 times its width. The function $y = (45w)(w)$ represents this situation, where y is the area, in square meters, of the rectangle and $y > 0$. Which of the following is the best interpretation of $45w$ in this context?

A) The length of the rectangle, in meters
B) The area of the rectangle, in square meters
C) The difference between the length and the width of the rectangle, in meters
D) The width of the rectangle, in meters

Question 12

Skill: Function Notation & Evaluation | Difficulty: Medium

$f(x) = 37$

For the given linear function f, which table gives three values of x and their corresponding values of f(x)?

A) x: 0, 1, 2 → f(x): 0, 0, 0
B) x: 0, 1, 2 → f(x): 37, 37, 37
C) x: 0, 1, 2 → f(x): 0, 37, 74
D) x: 0, 1, 2 → f(x): 37, 0, –37

Question 13

Skill: Lines & Angles | Difficulty: Medium

[Figure: Two lines r and s are intersected by a transversal line k. At the intersection with line r, angles w° and x° are formed. At the intersection with line s, angles y° and z° are formed. Note: Figure not drawn to scale.]

In the figure shown, line k intersects lines r and s. If $w = 145$, which additional piece of information is sufficient to prove that lines r and s are parallel?

A) $x = 35$
B) $y = 145$
C) $w + y = 180$
D) $y + z = 180$

Question 14

Skill: Circle Equations | Difficulty: Hard

$x^2 - 6x + y^2 - 10y - 66 = 0$

In the xy-plane, the graph of the given equation is a circle. If this circle is inscribed in a square, what is the perimeter of the square?

A) 20
B) 40
C) 80
D) 320

Question 15

Skill: Ratios, Rates & Proportions | Difficulty: Hard

A fitness studio offers a series of three bootcamp classes. 800 people attended the first class. 55% of the people who attended the first class attended the second class, and 40% of the people who attended the first and second classes attended the third class. How many people attended all three classes?

Enter your answer as a number.

Question 16

Skill: Exponential Functions - Growth & Decay | Difficulty: Hard

$f(x) = 30(3)^{x/5}$

Which table gives four values of x and their corresponding values of f(x) for the given exponential function?

A) x: –5, 0, 5, 10 → f(x): 10, 0, 30, 60
B) x: –5, 0, 5, 10 → f(x): 10, 30, 90, 270
C) x: –5, 0, 5, 10 → f(x): –10, 30, 90, 270
D) x: –5, 0, 5, 10 → f(x): 10, 30, 60, 90

Question 17

Skill: Volume & Surface Area | Difficulty: Hard

A cylindrical water tank has a radius of 5 meters and a height of 8 meters. If the tank is filled to 75% of its capacity, what is the volume of water in the tank, in cubic meters? (Use $\pi \approx 3.14$)

Enter your answer as a number.

Question 18

Skill: Volume & Surface Area | Difficulty: Hard

To study fluctuations in mineral content, samples of rock were taken from 18 quarries, each cut in the shape of a cube. The length of the edge of one of these cubes is 4.000 centimeters. This cube has a density of 0.500 grams per cubic centimeter. What is the mass of this cube, in grams?

Enter your answer as a number.

Question 19

Skill: Equivalent Expressions & Algebraic Manipulation | Difficulty: Hard

$-3|4x + 6| + 9 = -9$

What are all solutions to the given equation?

A) 0
B) 0 and –3
C) 0 and –14/4
D) There is no solution.

Question 20

Skill: Quadratic Equations - Solving | Difficulty: Hard

$x^2 - 72x - 18 = 0$

What is the sum of the solutions to the given equation?

A) 0
B) 9
C) 18
D) 72

Question 21

Skill: Volume & Surface Area | Difficulty: Hard

[Figure: A right circular cone with apex A at the top, and B is a point on the circumference of the base. The height of the cone is 12 centimeters and the slant height (segment AB, not shown) is 20 centimeters. Note: Figure not drawn to scale.]

For the right circular cone shown, B is a point on the circumference of the base, and the length of segment AB (not shown) is 20 centimeters. If the height of the cone is 12 centimeters and the volume of the cone is $k\pi$ cubic centimeters, what is the value of k?

Question 22

Skill: Coordinate Geometry | Difficulty: Hard

In the xy-plane, line j and line m are perpendicular and intersect at the point (4, 6). If line j is defined by the equation $y = mx + b$, where m and b are constants and $m > 1$, which of the following points lies on line m?

A) (5, 6 – 1/m)
B) (5, 6 + 1/m)
C) (5, 6 – m)
D) (5, 6 + m)

Section 2: Math

Module 2 - Easier Path (22 Questions)

Time: 35 minutes

Question 1

Skill: Quadratic Functions - Graphs & Properties | Difficulty: Easy

[Graph: A parabola opening downward showing the height above ground (in meters) of a ball x seconds after the ball was launched upward from a platform. The graph passes through approximately (0, 3), peaks near (0.8, 5.2), and returns to 0 near (1.8, 0). A point is marked at (1.2, 4.5).]

The graph shows the height above ground, in meters, of a ball x seconds after the ball was launched upward from a platform. Which statement is the best interpretation of the marked point (1.2, 4.5) in this context?

A) 1.2 seconds after being launched, the ball's height above ground is 4.5 meters.

B) 4.5 seconds after being launched, the ball's height above ground is 1.2 meters.

C) The ball was launched from an initial height of 1.2 meters with an initial velocity of 4.5 meters per second.

D) The ball reaches its maximum height of 4.5 meters at 1.2 seconds.

Question 2

Skill: Linear Equations in One & Two Variables | Difficulty: Easy

In one month in 2019, a freelancer earned a total of $900 by working at a primary contract and at a secondary contract doing part-time work. The equation $20h + 15c = 900$ represents this situation, where h is the number of hours worked at the primary contract and c is the number of hours worked at the secondary contract. Which of the following is the best interpretation of 20 in this context?

A) The number of hours the freelancer worked at the primary contract

B) The amount, in dollars, the freelancer earned for each hour at the secondary contract

C) The amount, in dollars, the freelancer earned for each hour at the primary contract

D) The number of hours the freelancer worked at the secondary contract

Question 3

Skill: Standard Deviation | Difficulty: Easy

Which of the following lists represents a data set with the smallest standard deviation?

A) 30, 31, 33, 35, 36

B) 31, 31, 33, 35, 35

C) 31, 32, 33, 34, 35

D) 32, 33, 33, 33, 34

Question 4

Skill: Similar Triangles | Difficulty: Medium

[Figure: In the figure shown, triangle FGH is similar to triangle FIJ. The measure of angle FIJ is 62 degrees, and FH = 18(IJ). Note: Figure not drawn to scale.]

In the figure shown, triangle FGH is similar to triangle FIJ. The measure of angle FIJ is 62 degrees, and FH = 18(IJ). What is the measure of angle FGH?

A) 18°
B) 62°
C) (18 + 62)°
D) (18 × 62)°

Question 5

Skill: Probability - Basic | Difficulty: Medium

At a convention center, there are a total of 500 attendees. Each attendee is located in either Hall X, Hall Y, or Hall Z. If one of these attendees is selected at random, the probability of selecting an attendee in Hall X is 0.54, and the probability of selecting an attendee in Hall Y is 0.36. How many attendees are located in Hall Z?

A) 10
B) 180
C) 90
D) 50

Question 6

Skill: Systems of Linear Equations | Difficulty: Medium

$y = x/3 + 8$
$y = -x/3 + 16$

The solution to the given system of equations is (x, y). What is the value of 2y?

A) 24
B) 20
C) 16
D) 8

Question 7

Skill: Linear Equations in One & Two Variables | Difficulty: Easy

In 7 days, a hummingbird consumed 31.5 grams of nectar. Which equation describes the amount of nectar y, in grams, the hummingbird consumed in these 7 days?

A) y = 31.5 + 7
B) y = 31.5
C) y = 31.5/7
D) y = 7

Question 8

Skill: Exponential Functions - Growth & Decay | Difficulty: Medium

A pharmacologist observes a sample of a drug compound. An exponential model estimates that the mass, in milligrams, of the sample decreases by 18% every 8.5 minutes. Which of the following equations could represent this model, where M is the estimated mass, in milligrams, of the sample t minutes after the pharmacologist began observing?

A) $M = 50(0.18)^{t+8.5}$
B) $M = 50(0.18)^{t/8.5}$
C) $M = 50(0.82)^{t+8.5}$
D) $M = 50(0.82)^{t/8.5}$

Question 9

Skill: Coordinate Geometry | Difficulty: Medium

Line p is defined by y = –5x + 9. Line q is parallel to line p in the xy-plane and passes through the point (0, 12). Which equation defines line q?

A) y = 12x + 9
B) y = –12x + 9
C) y = 5x + 12
D) y = –5x + 12

Question 10

Skill: Function Notation & Evaluation | Difficulty: Hard

For the linear function g, g(c) = –4, where c is a constant, g(3) = 32, and the slope of the graph of y = g(x) in the xy-plane is 6. For the linear function h, h(c) = –5 and h(6) = 49. What is the slope of the graph of y = h(x) in the xy-plane?

A) –1
B) 2
C) 6
D) 8

Question 11

Skill: Polynomial Operations | Difficulty: Hard

If 3(5 – 2x) + 8(5 – 2x) + 4 = 7(5 – 2x) + 10, what is the value of 2x – 5?

Enter your answer as a number.

Question 12

Skill: Linear Inequalities | Difficulty: Hard

[Graph: A coordinate plane showing a shaded region above a line. The boundary line passes through approximately (–8, –8) and (0, –6), with a positive slope of 0.25. The region above and to the left is shaded.]

The shaded region shown represents the solutions to rx + ty ≥ –48, where r and t are constants. What is the value of r + t?

A) –5
B) –2
C) 2
D) 6

Question 13

Skill: Radical and Rational Exponents | Difficulty: Hard

$f(x) = \sqrt{3x + 2}$

The function f is defined by the given equation. If $f(a) = –3a$, where a is a constant, what is the value of a?

A) 1/3
B) 2/9
C) –2/9
D) –1/3

Question 14

Skill: Polynomial Operations | Difficulty: Hard

$\sqrt[4]{p^6} = t$

In the given equation, $p > 1$ and $t > 1$. If $t = p^{2n–1}$, where n is a constant, what is the value of n?

Enter your answer as a fraction.

Question 15

Skill: Quadratic Functions - Graphs & Properties | Difficulty: Hard

The function g is a quadratic function. In the xy-plane, the graph of $y = g(x)$ has a vertex at (1, 4) and passes through the points (2, 38) and (–1, 140). What is the value of $g(–2) – g(0)$?

A) 108
B) 176
C) 272
D) 312

Question 16

Skill: Quadratic Functions - Graphs & Properties | Difficulty: Hard

$f(x) = –2(x – 3)(x + 7)$

For the given function, what is the y-coordinate of the y-intercept of the graph of $y = f(x)$ in the xy-plane?

Enter your answer as a number.

Question 17

Skill: Exponential Functions - Growth & Decay | Difficulty: Hard

[Graph: The graph of y = f(x) + 3 is shown. The curve passes through approximately (0, 5) and decreases steeply as x increases, crossing y = 0 near x = 1.2. As x decreases toward negative infinity, the curve approaches y = 6 from below.]

The graph of $y = f(x) + 3$ is shown. Which equation defines function f?

A) $f(x) = –4^x + 1$

B) $f(x) = -4^x + 3$
C) $f(x) = -4^x + 5$
D) $f(x) = -4^x + 7$

Question 18

Skill: Ratios, Rates & Proportions | Difficulty: Hard

A square blueprint has a side length of 60 inches, and 1 inch on the blueprint represents an actual distance of 8 miles. A smaller version of the same blueprint is printed as a square with the side length 65% shorter than the side length of the previous blueprint. On the smaller blueprint, which of the following is closest to the actual distance, in miles, represented by 1 inch?

A) 5.20
B) 12.31
C) 22.86
D) 36.57

Question 19

Skill: Quadratic Equations - Solving | Difficulty: Hard

$3x^2(2x - 5)(2x - u) = 0$

In the given equation, u is a positive constant. The sum of the solutions to the equation is 17/4. What is the value of u?

Enter your answer as a fraction.

Question 20

Skill: Similar Triangles | Difficulty: Hard

Right rectangular prism P is similar to right rectangular prism Q. The surface area of rectangular prism P is 72 square centimeters (cm^2), and the surface area of rectangular prism Q is 1,800 cm^2. The volume of rectangular prism Q is 2,500 cubic centimeters (cm^3). What is the sum of the volumes, in cm^3, of right rectangular prism P and right rectangular prism Q?

Enter your answer as a number.

Question 21

Skill: Exponential Functions - Growth & Decay | Difficulty: Hard

The function g is defined by $g(x) = cd^{x/m}$, where c, d, and m are constants, and d and m are integers. If $g(3) = 8$ and $g(6) = 216$, what is the value of $g(9)$?

Enter your answer as a number.

Question 22

Skill: Quadratic Word Problems | Difficulty: Hard

A ball is thrown upward from a height of 6 feet. The height h, in feet, of the ball t seconds after it is thrown is given by $h = -16t^2 + 40t + 6$. How many seconds after being thrown does the ball reach its maximum height?

A) 0.75
B) 3.00
C) 2.50
D) 1.25

Section 2: Math

Module 2 - Harder Path (22 Questions)

Time: 35 minutes

Question 1

Skill: Systems of Linear Equations | Difficulty: Hard

$2x + 3y = 14$
$5x + 7y = 33$

The solution to the given system of equations is (x, y). What is the value of $x + y$?

A) 3
B) 5
C) 7
D) 9

Question 2

Skill: Systems of Linear Equations | Difficulty: Hard

A chemist has two solutions. Solution A is 30% acid and Solution B is 70% acid. How many liters of each solution must be mixed to produce 20 liters of a solution that is 55% acid?

How many liters of Solution A are needed? Enter your answer as a number.

Question 3

Skill: Systems of Equations - Number of Solutions | Difficulty: Hard

The equation $4(3x + k) = 12x + 20$ has infinitely many solutions. What is the value of k?

Enter your answer as a number.

Question 4

Skill: Function Notation & Evaluation | Difficulty: Hard

The function f is defined by $f(x) = 2x^2 + bx + c$, where b and c are constants. In the xy-plane, the graph of $y = f(x)$ has a vertex at $(3, -7)$. What is the value of $f(0)$?

A) 11
B) 7
C) –7
D) 25

Question 5

Skill: Function Notation & Evaluation | Difficulty: Hard

For a function f, $f(x) = (x + 3)^2 - 16$. The function g is defined as $g(x) = f(x + 2)$. For what value of x does $g(x)$ reach its minimum?

Enter your answer as a number.

Question 6

Skill: Function Notation & Evaluation | Difficulty: Hard

Let $h(x) = ax + b$, where a and b are constants. If $h(h(x)) = 9x + 20$, what is the value of $a + b$?

A) 5
B) 7
C) 8
D) 10

Question 7

Skill: Polynomial Operations | Difficulty: Hard

If $(x + y)^2 = 49$ and $xy = 10$, what is the value of $x^2 + y^2$?

A) 19
B) 59
C) 39
D) 29

Question 8

Skill: Polynomial Operations | Difficulty: Hard

If $27^x = 9^{y+1}$ and $x + y = 5$, what is the value of x?

Enter your answer as a number.

Question 9

Skill: Polynomial Operations | Difficulty: Hard

When the polynomial $p(x) = 2x^3 + ax^2 + bx + 6$ is divided by $(x – 1)$, the remainder is 3. When p(x) is divided by $(x + 1)$, the remainder is 5. What is the value of a?

Enter your answer as a number.

Question 10

Skill: Quadratic Equations - Solving | Difficulty: Hard

The equation $x^2 + kx + 36 = 0$, where k is a positive constant, has exactly one solution. What is the value of k?

A) 6
B) 12
C) 18
D) 36

Question 11

Skill: Quadratic Word Problems | Difficulty: Hard

The function $f(x) = -x^2 + 10x - 16$ models the profit, in thousands of dollars, of a company that sells x thousand units of a product. What is the maximum profit the company can achieve, in thousands of dollars?

Enter your answer as a number.

Question 12

Skill: Quadratic Equations - Solving | Difficulty: Hard

The sum of two numbers is 20 and their product is 96. What is the positive difference between the two numbers?

A) 2
B) 12
C) 8
D) 4

Question 13

Skill: Triangle Properties | Difficulty: Hard

In triangle ABC, angle A = 90 degrees, AB = 5, and BC = 13. Point D lies on segment AC such that BD is perpendicular to AC. What is the length of BD?

Enter your answer as a fraction.

Question 14

Skill: Circle Equations | Difficulty: Hard

A circle has center (h, k) and radius r. The circle passes through the points (2, 6), (8, 6), and (2, 0). What is the value of r^2?

A) 9
B) 18
C) 25
D) 36

Question 15

Skill: Volume & Surface Area | Difficulty: Hard

A hemisphere with radius 6 cm is placed on top of a cylinder with radius 6 cm and height 10 cm. What is the total volume of the combined solid, in cubic centimeters? (Express your answer in terms of π.)

Enter your answer in terms of π.

Question 16

Skill: Coordinate Geometry | Difficulty: Hard

In the xy-plane, a line passes through the points (2, 5) and (6, 1). What is the area of the triangle formed by this line and the positive x-axis and positive y-axis?

A) 12.25
B) 14.5
C) 24.5
D) 49

Question 17

Skill: Coordinate Geometry | Difficulty: Hard

In the xy-plane, the graph of $y = (x - 3)^2 + 2$ is reflected over the line $y = 2$. Which of the following equations describes the reflected graph?

A) $y = -(x - 3)^2 + 2$
B) $y = (x - 3)^2 - 2$
C) $y = -(x - 3)^2 + 4$
D) $y = (x + 3)^2 + 2$

Question 18

Skill: Probability - Basic | Difficulty: Hard

A data set contains 10 values. The mean of the data set is 24. When a new value of 57 is added to the data set, what is the new mean?

A) 24
B) 27
C) 30
D) 33

Question 19

Skill: Probability - Basic | Difficulty: Hard

A bag contains 5 red marbles, 4 blue marbles, and 3 green marbles. Two marbles are drawn at random without replacement. What is the probability that both marbles are blue?

A) 16/144
B) 1/9
C) 4/33
D) 1/11

Question 20

Skill: Exponential Functions - Growth & Decay | Difficulty: Hard

A population of bacteria doubles every 3 hours. If the initial population is 500, after how many hours will the population first exceed 10,000?

Enter your answer as a whole number.

Question 21

Skill: Right Triangle Trigonometry | Difficulty: Hard

In right triangle PQR, angle Q = 90 degrees. If sin(P) = 5/13, what is the value of cos(R)?

A) 5/13
B) 12/13
C) 5/12
D) 13/5

Question 22

Skill: Equivalent Expressions & Algebraic Manipulation | Difficulty: Hard

If $|2x - 3| = |x + 5|$, what is the sum of all possible values of x?

Enter your answer as a fraction.